

BEST PRACTICES IN ACCESSIBLE ONLINE DESIGN



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Preface PrefaceEdit Update titlecancel

Accessibility is frequently the last thing course and website developers want to think about when creating an online content. There is extra time involved up front, but it can help prevent problems down the line. I think most of us in higher education care about all students and want to help them to our best ability. There are also laws protecting the rights of individuals with disabilities to have access to programs and services that institutions of higher education offer. [Section 504 of The Rehabilitation Act of 1973](#) says that public institutions receiving federal funds, this includes student financial aid, need to ensure that people with disabilities can participate in programs & activities and have the same benefits that people without disabilities have. It requires academic adjustments and accommodation to ensure full participation. [Section 508](#) is an amendment that requires electronic information and technology, such as websites and online courses be accessible. [The American with Disabilities Act of 1990](#) expands the Rehabilitation Act of 1973 to private as well as public institutions of higher education. The newest is [The 21st Century Communications and Video Accessibility Act of 2010](#). It requires modern communications to be accessible to people with disabilities. This includes VOIP services, electronic messaging, video conferencing, video communications and mobile browsers.

One only needs to search Google to find names of institutions of higher education that have been sued for not having accessible online content and services. When we design accessibly from the start, we can attract a wider audience to our product or service, and protect our institutions from losing time and money in a future lawsuit.

I'm writing this book to help faculty and online site developers learn about best practices for designing and developing online content. Some of the instructions for document accessibility are based off of Office 2016, rather than Office 365 in browser apps. The menus should be similar.

This book is currently under development. So, please check regularly to see updates.

Thank you for your interest in making your content accessible and for caring about others!

-Heather Caprette

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1.1: Understanding Accessibility

1.1: Understanding Accessibility

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Accessibility is the degree to which a product, service or environment is available to all users, including those with disabilities or special needs. It's about our ability to participate in and belong to the world around us. The Office of Civil Rights, in a resolution agreement with South Carolina Technical College, said, "Accessible means a person with a disability is afforded the opportunity to acquire the same information, engage in the same interactions, and enjoy the same services as a person without a disability in an equally effective and equally integrated manner, with substantially equivalent ease of use."

A similar concept is Universal Design. Universal Design is the proactive design of products and environments to be usable by all people, to the greatest extent, without the need for adaptation or specialized design. A curb cut is an example of something that is universally designed. It helps people in wheel chairs and using walkers navigate their environment, but it also benefits people pushing strollers and carts, pulling luggage, people on bicycles, and people on roller skates.



Curb cuts are an example of universal design. They benefit everyone, not just people who use wheelchairs.

While it may not be possible to make everything accessible to all people due to variation in user needs and abilities, we can follow standards to make our online content accessible to the broadest range of people. These standards are referred to as the Web Content Accessibility Guidelines.

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1.2: What is Disability 1.2: What is DisabilityEdit Update titlecancel

According to [U.S. Code Title 42, Chapter 126, § 12102](#), a disability is a physical or mental impairment that substantially limits one or more major life activities of an individual. Major life activities include, but are not limited to, caring for oneself, performing manual tasks, seeing, hearing, eating, sleeping, walking, standing, lifting, bending, speaking, breathing, learning, reading, concentrating, thinking, communicating, and working.

A major life activity also includes the operation of a major bodily function, including but not limited to, functions of the immune system, normal cell growth, digestive, bowel, bladder, neurological, brain, respiratory, circulatory, endocrine, and reproductive functions.

In reality, we are all aging into disability.

Visual disabilities include blindness, low vision, tunnel vision, and central field loss in which there is loss of vision in the central portion of the eye. Auditory disabilities include deafness, hearing impairment and speech disabilities. Diseases such as Parkinson's and arthritis can affect mobility, use of hands, eyes, coordination and fine motor control. Cognitive disabilities are highly variable and can include learning disability, dyslexia, and conditions that affect memory and attention.

Sometimes when people think of a person with a disability, they think of a condition that has existed since birth. Not all disabilities are present at birth and not all are permanent. A person may lose the use of his or her dominant arm or hand for a few months due to a fractured or broken bone, for example. A disability can happen to anyone of us and anytime.

Online we will be concerned with the accessibility of our documents, such as Word, PDF and Power Point files, the code behind what we present, the websites we ask our users to interact with, and the activities we ask them to perform.

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1.3: Person First Language 1.3: Person First LanguageEdit Update titlecancel

When I was in Physical Therapy school at OSU, I had an instructor tell us that we have to be careful not to refer to the patient by their condition. That is, you wouldn't want to say, "I have my hip replacement coming at 1 o'clock." Instead you might tell your supervisor or fellow physical therapist, "I have Mrs. M coming at 1 o'clock for her hip replacement therapy." Likewise, when we have a student with a disability, such as blindness, we would refer to him or her as a person who is blind. For example, if I were to approach our Office of Disability Services, I would say, "I need help with making accommodations for a student in my course who is blind," instead of "I need help with making accommodations for a blind student in my course."

The Center for Disease Control has a [helpful chart of people first language](#) and language to avoid. They've put this language in a poster called [Communicating With and About People With Disabilities](#) (visit: https://www.cdc.gov/ncbddd/disabilit...ter_photos.pdf)

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<https://pressbooks.ulib.csuohio.edu/accessibility/?p=635>

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1.4: Why Should We Design for Accessibility

1.4: Why Should We Design for Accessibility

Please view the video titled “[Keeping Web Accessibility In Mind](#)”

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[Transcript for Keeping Web Accessibility In Mind.](#)

As teachers and members of an academic institution, don’t we want everyone to have the opportunity to learn, share, be creative and succeed in life? Aren’t we here to enrich people’s lives and provide information to everyone? Imagine if we can open doors to information, not shut them to a certain segment of the population. We want to enable independence for people with disabilities.

To quote David Berman, a web accessibility expert, “what if we had denied an education to these people?”



Ron Mace is known as the father of universal design. He became an architect and worked on the first U.S. building codes to incorporate principles of accessible design, in North Carolina in 1973. In college, he was carried into buildings that had no wheelchair ramps.



Stephen Hawking is a famous mathematician and physicist, teaching at Cambridge University. He wrote “A Short History of the Universe” and served as the prestigious Lucasian Professor of Mathematics at Cambridge.



Helen Keller was the first blind and deaf person to receive a Bachelor of Arts degree. She became a lecturer, writer, and political activist.

If you've ever experienced being excluded from an activity you wanted to participate in, you'll understand, that designing inclusive experiences is the right thing to do.

It's smart for business, helping users find our information and broadening our customer base.

Moreover, it's the law.

For another perspective on designing for accessibility, view David Berman's video titled "[Why Should We Care.](#)"

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1.5: The Laws Supporting Accessibility 1.5: The Laws Supporting AccessibilityEdit Update titlecancel

There are four existing laws you should be aware of when designing online content at an institution of higher education.

1. **Section 504 of the Rehabilitation Act of 1973**, says that colleges or universities that receive federal funds need to ensure that people with disabilities can participate in programs & activities and have the same benefits that people without disabilities have. Subpart E requires an institution to be prepared to make reasonable academic adjustments and accommodations to allow students with disabilities full participation in the same programs and activities available to students without disabilities.
2. **Section 508 of the Rehabilitation Act of 1973** is an amendment that deals with electronic information and technology. If you are a state that receives federal funds under the Technology Related Assistance for Individuals with Disabilities Act of 1988, you need to ensure that people with disabilities have access to and can use information and services available to non-disabled members of the public. This includes information stored on web sites and in online courses. Ohio receives funds from the Technology Related Assistance for Individuals with Disabilities Act of 1988. With a recent rewriting of Section 508, on January 18, 2017, the rule now refers to Electronic and Information Technology (EIT) as Information and Communication Technology (ICT).
3. **The American with Disabilities Act of 1990** expands the reach of the Rehabilitation Act of 1973 to private as well as public institutions of higher education. It requires schools to provide access to the same programs that non-disabled people participate in, in an integrated setting where possible. It requires auxiliary aids and services when necessary to ensure effective communication.
4. **The 21st Century Communications and Video Accessibility Act of 2010** requires modern communications to be accessible to people with disabilities. This means VOIP services, electronic messaging, video conferencing, video communications & mobile browsers.

For those who are curious, the following sites have lists of higher education accessibility lawsuits, complaints and settlements:

1. [University of Minnesota Duluth's list of lawsuits](#)
2. [University of Washington's page of resolution agreements and lawsuits](#)

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1.6: A Shift Toward Broader Standards and Functionality Supporting Accessible Online Content

1.6: A Shift Toward Broader Standards and Functionality Supporting Accessible Online Content

Edit Update titlecancel

Legislation regarding accessibility at higher educational institutions recently changed. In the United States, we follow Section 508 guidelines for designing websites and online content. The U.S. Access Board writes Section 508 rules for organizations receiving federal funds to follow, as part of the 1998 Amendment to the Rehabilitation Act. They are enforced by the Dept. of Education's Office of Civil Rights and the Department of Justice.

Section 508 rules were previously published on December 21, 2000, and implemented in the summer of 2001. However, Section 508 recently underwent a refresh, and the new standards were written into the Federal Registrar on January 18, 2017. The new standards align with the Web Content Accessibility Guidelines 2.0 (WCAG 2.0) Level AA. At the time of writing, WCAG went through an update to version 2.1 on June 5th, 2018. We'll have to keep checking to see when Section 508 points to the newer standards. The Web Content Accessibility Guidelines are written by the Web Accessibility Initiative (WAI) at the World Wide Web Consortium. This is an international group and these are international standards that have been adopted by designers in other countries, such as in Europe and Canada, already. There are three levels to each WCAG 2.0 guideline – Level A is what you must do, level AA is what you should do, and level AAA produces the best accessibility for the most people. The WCAG 2.0 guidelines are broader in scope than the previous Section 508 standards. The new focus will be on functionality rather than device or technology specific standards. Technology has changed a great deal since Section 508 standards were last written in 2000. People are no longer relying on desktops primarily to access information. They are using smart phones and other mobile devices with voice text and video communications. By harmonizing U.S. standards with what other countries are following, we will create a larger market for our information and communication technology. Along with the Section 508 rules rewrite, section 255 of the Communications Act of 1934 is also being updated. Section 255 applies to the manufacturers and authors of our communication and information technology devices and software. The information and communication technology the new standards refer to are: VOIP products, computers, software, office machines, information kiosks, transaction machines, websites, and electronic documents.

The image below shows the degree of accessibility each level of WCAG 2.0 covers compared to the previous Section 508 standards, which most closely resembles WCAG 2.0 Level A.



Another change to Section 508 is delineation of covered “content.” All public facing content as well as eight categories of non-public facing content will have to be accessible.

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